

Triage and the "ABCD" concept

Many deaths in hospital occur within 24 hours of admission. Some of these deaths can be prevented if very sick children are quickly identified on their arrival and treatment is started without delay. In many hospitals around the world, children are not checked before a senior health worker examines them; as a result, some seriously ill patients have to wait a very long time before they are seen and treated. Children are known to have died of a treatable condition when waiting in the queue for their turn. The idea of triage is to prevent this from happening. The word “triage” means sorting. The use of triage to prioritize the critically ill dates back to the early 19th century, when this was developed by military surgeons in the Napoleonic war between France and Russia.

Triage is the process of rapidly examining all sick children when they first arrive in hospital in order to place them in one of the following categories:

- Those with **EMERGENCY SIGNS** who require immediate emergency treatment.
If you find any emergency signs, do the following **immediately**:
 - Start to give appropriate emergency treatment.
 - Call a senior health worker and other health workers to help.
 - Carry out emergency laboratory investigations.
- Those with **PRIORITY SIGNS**, indicating that they should be given priority in the queue, so that they can **rapidly** be assessed and treated without delay.
- Those who have no emergency or priority signs and therefore are **NON-URGENT** cases. These children can wait their turn in the queue for assessment and treatment.¹ The majority of sick children will be non-urgent and will not require emergency treatment.

After these steps are completed, proceed with general assessment and further treatment according to the child's priority.

In an ideal situation, all children should be checked on their arrival in hospital by a person who is trained to assess how ill they are. This person decides whether the patient will be seen immediately and will receive life-saving treatment, or will be seen soon, or can safely wait his/her turn to be examined.

TRIAGE

is the sorting of patients into priority groups according to their need and the resources available

E	Emergency
P	Priority
Q	Queue (non-urgent)

¹ Sometimes it is discovered that a child in the queue is waiting for immunization. These children do not need assessment and can be referred to the right department without delay.

Categories after triage	Action required
EMERGENCY CASES	Need immediate emergency treatment
PRIORITY CASES	Need assessment and rapid attention
NON-URGENT CASES	Can wait their turn in the queue

A	Airway
B	Breathing
C	Circulation
	Coma
	Convulsion
D	Dehydration (severe)

The ABCD concept

Triage of patients involves looking for signs of serious illness or injury. These emergency signs relate to the Airway-Breathing-Circulation/Consciousness-Dehydration and are easily remembered as “**ABCD**”.

Each letter refers to an emergency sign which, when positive, should alert you to a patient who is seriously ill and needs immediate assessment and treatment.

Priority signs

Besides the group of emergency signs described above, there are priority signs, which should alert you to a child who needs prompt, but not emergency assessment. These signs can be remembered with the symbols **3 TPR - MOB**:

- Tiny baby: any sick child aged under two months
- Temperature: child is very hot
- Trauma or other urgent surgical condition
- Pallor (severe)
- Poisoning
- Pain (severe)
- Respiratory distress
- Restless, continuously irritable, or lethargic
- Referral (urgent)
- Malnutrition: Visible severe wasting
- Oedema of both feet
- Burns

The frequency with which children showing some of these priority signs appear in the outpatient department depends on the local epidemiology. The signs might need to be adapted accordingly, for example by including signs for common severe conditions which cannot wait in your setting.

The triaging process

Triaging should not take much time. For a child who does not have emergency signs, it takes on average 20 seconds. The health worker should learn to assess several signs at the same time. A child who is smiling or crying does not have severe respiratory distress, shock or coma. The health worker looks at the child, observes the chest for breathing and priority signs such as severe malnutrition and listens to abnormal sounds such as stridor or grunting.

Several methods are available to facilitate the triaging process. One example is a stamp being used in Malawi consisting of the “ABCD” signs in which the health worker circles the correct step and initiates **emergency** treatment “E” or puts them in **priority** groups “P” or “Q” for children who can wait in the **queue**.

Colours can also be used for differentiating the three groups, giving a red sticker to emergency cases, a yellow for priority and green for the queue.

A	Airway
B	Breathing
C	Circulation
Cm	Coma
Cn	Convulsion
D	Dehydration (severe)

WHEN AND WHERE SHOULD TRIAGING TAKE PLACE?

Triage should be carried out as soon as a sick child arrives in the hospital, well before any administrative procedure such as registration. This may require reorganizing the flow of patients in some locations.

Triage can be carried out in different locations – e.g. in the outpatient queue, in the emergency room, or in a ward if the child has been brought directly to the ward at night. In some settings, triage is done in all these places. Emergency treatment can be given wherever there is room for a bed or trolley for the sick child and enough space for the staff to work on the patient, and where appropriate drugs and supplies are easily accessible. If a child with emergency signs is identified in the outpatient queue, he/she must quickly be taken to a place where treatment can be provided immediately, e.g. the emergency room or ward.

E	Emergency
P	Priority
Q	Queue

WHO SHOULD TRIAGE?

All clinical staff involved in the care of sick children should be prepared to carry out rapid assessment in order to identify the few who are severely ill and require emergency treatment. If possible, all such staff should be able to give initial emergency treatment, as described in the flowchart and treatment charts. In addition, people such as gatemen, record clerks, cleaners, janitors who have early patient contact should be trained in triage for emergency signs and should know where to send people for immediate management.

HOW TO TRIAGE?

Keep in mind the ABCD steps: Airway, Breathing, Circulation, Coma, Convulsion, and Dehydration.

To assess if the child has airway or breathing problems you need to know:

- Is the child breathing?
- Is the airway obstructed?
- Is the child blue (centrally cyanosed)?

Look, listen and feel for air movement. Obstructed breathing can be due to blockage by the tongue, a foreign body, a swelling around the upper airway (retropharyngeal abscess) or severe croup which may present with abnormal sounds such as stridor.

- Does the child have severe respiratory distress?

Is the child having trouble getting breath so that it is difficult to talk, eat or breastfeed? Is he breathing very fast and getting tired, does he have severe chest indrawing or is he using auxiliary respiratory muscles?

To assess if the child has circulation problems you need to know:

- Does the child have warm hands?
- If not, is the capillary refill time longer than 3 seconds?
- And is the pulse weak and fast?

In the older child the radial pulse may be used; however, in the infant, the brachial or femoral pulses may need to be felt.

To assess for coma you need to know:

A rapid assessment of conscious level can be made by assigning the patient to one of the AVPU categories:

A	A lert
V	responds to V oice
P	responds to P ain
U	U nresponsive

A child who is not alert but responds to voice is lethargic. If the assessment shows that the child does not respond to voice and only responds to pain (with targeted or untargeted movements), or does not respond at all, the level is at "P" or "U". We then refer to that child as having coma and the child needs to be treated accordingly.

To assess for dehydration you need to know:

- If the child is lethargic or unconscious
- If the child has sunken eyes
- If the skin pinch goes back very slowly

When ABCD has been completed and there are no emergency signs, continue to assess the priority signs.

If the child has **any sign** of the ABCD, it means the child has an emergency "**E**" sign and **emergency treatment** should start **immediately**

Assessing priority signs

If the child does not have any of the E signs, the health worker proceeds to assess the child on the priority signs. This should not take more than few seconds. Some of these signs will have been noticed during the ABCD triage discussed so far, and others need to be rechecked. Follow the 3 TPR-MOB to quickly complete this section.

Tiny infant (less than two months of age)

If the child appears very young, ask the mother his age. If the child is obviously not a young infant, you do not need to ask this question.

Small infants are more difficult to assess properly, more prone to getting infections (from other patients), and more likely to deteriorate quickly if unwell. All tiny babies of under two months should therefore be seen as a priority.

Temperature: Hot (fever - high Temperature)

A child that feels very hot may have high fever. Children with high fever on touch need prompt treatment. Take the waiting child to the front of the queue and take locally adopted action, like having the temperature checked by thermometer, giving an antipyretic, or doing investigations like a blood film for malaria.

Severe Trauma (or other urgent surgical condition)

Usually this is an obvious case, but one needs to think of acute abdomen, fractures and head injuries in this category.

Severe Pallor

Pallor is unusual paleness of the skin, and severe pallor is a sign of severe anaemia which might need urgent transfusion. It can be detected by comparing the child's palms with your own. If the palms are very pale (almost paper-white), the child is severely anaemic.

Poisoning

A child with a history of swallowing drugs or other dangerous substances needs to be assessed immediately, as he can deteriorate rapidly and might need specific treatments depending on the substance taken. The mother will tell you if she has brought the child because of possible intoxication.

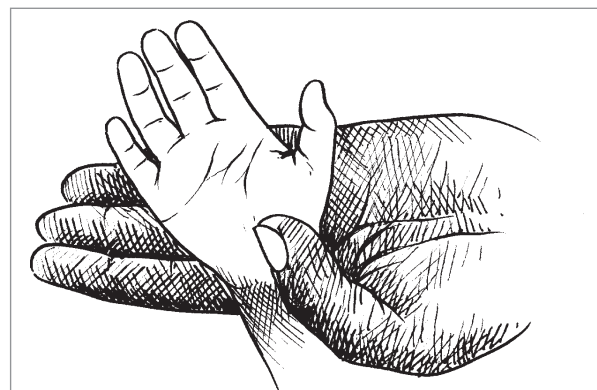


Figure 1
Severe pallor

Severe Pain

If a child has severe pain and is in agony, she/he should be prioritized to receive early full assessment and pain relief. Severe pain may be due to severe conditions such as acute abdomen, meningitis, etc.

Lethargy or Irritable and Restless

Recall from your assessment of coma with the AVPU scale whether the child was lethargic. A lethargic child responds to voice but is drowsy and uninterested (V in the AVPU scale).

The continuously irritable or restless child is conscious but cries constantly and will not settle.

Respiratory distress

When you assessed the airway and breathing, did you observe any respiratory distress? If the child has severe respiratory distress, it is an emergency. There may be signs present that you do not think are severe, e.g. lower chest wall indrawing (not severe), or difficulty in breathing. In this case, the child does not require emergency treatment but will need urgent assessment. Decisions on the severity of respiratory distress come with practice. If you have any doubts, have the child seen and treated immediately.

Urgent Referral

The child may have been sent from another clinic. Ask the mother if she was referred from another facility and for any note that may have been given to her. Read the note carefully and determine if the child has an urgent problem.

Severe wasting (Severe Malnutrition)

A child with visible severe wasting has a form of malnutrition called marasmus. To assess for this sign, look rapidly at the arms and legs as well as the child's chest. A detailed description is given in the section on dehydration (see pages 28/29).

Oedema of both feet

Oedema of both feet is an important diagnostic feature of kwashiorkor, another form of severe malnutrition. Other signs are changes in the skin and hair.

Major Burn

Burns are extremely painful and children who seem quite well can deteriorate rapidly. If the burn occurred recently, it is still worthwhile to cool the burnt area with water, for example, by sitting the child in a bathtub with cool water. Any child with a major burn, trauma or other surgical condition needs to be seen quickly. Get surgical help or follow surgical guidelines.

Triage all sick children. When a child with emergency signs is identified, take to the emergency room or treatment area and start the appropriate emergency treatments immediately. Do not proceed to the next step before treatment is begun for a positive sign.

Treatment begins
when
any emergency
sign is identified

Triage Steps	Treat when any sign is positive
Assess A	If positive, treat. If negative, proceed to B
Assess B	If positive, treat. If negative, proceed to C
Assess C	If positive, treat. If negative, proceed to D
Assess D	If positive, treat. If negative, proceed to priority signs

If the child has no emergency signs, check for priority signs. After the examination for priority signs has been completed, the child will be assigned to one of:

- Priority (P):** the child should be put at the front of the queue
Queue (Q): if the child has no emergency or priority signs

Once appropriate emergency treatments have been initiated:

- call for a senior health worker;
- draw blood for emergency laboratory investigations such as blood glucose, haemoglobin and malaria smear;
- ask about head or neck trauma before providing treatment;
- take careful note if the child is severely malnourished, because this will affect the treatment of shock and dehydration caused by diarrhoea.

It is essential to act as quickly as possible and to start the emergency treatments. The team needs to stay calm and work together efficiently. The person in charge of the emergency management of the child assigns tasks so that the assessment can continue and treatments can be initiated as quickly as possible. Other health workers help to give the treatment needed, especially since a very sick child may need several treatments at once. The senior health worker will direct the treatment and immediately continue with a core assessment and follow-up of the child, identifying all the child's problems and developing the treatment plan.

More than one
treatment may have to
be administered as
quickly as possible, and
several people may
have to work together
as a team

GENERAL TREATMENT FOR PRIORITY SIGNS

Priority signs lead to quicker assessment of the child by moving the child to the front of the queue. While waiting, some supportive treatments may already be given. For example, a child found to have a hot body may receive an antipyretic

such as paracetamol. Similarly, a child with a burn may have severe pain and the pain could be controlled while waiting for definitive treatment.

If a child has no emergency signs or priority signs, she/he may return to the queue.

THE NEED FOR FREQUENT REASSESSMENT

During and following emergency treatment, the child should be re-assessed using the complete ABCD process. The disease course is dynamic and there could be new developments within a short time. Reassessment should begin with assessment of the airway and through the ABCD.

Triage is the sorting of patients into priority groups according to their need.

All children should undergo triage. The main steps in triage are:

- Look for emergency signs.
- Treat any emergency signs you find.
- Call a senior health worker to see any emergency.
- Look for any priority signs.
- Place priority patients at the front of the queue.
- Move on to the next patient.

Triage should be carried out quickly. You will soon learn to observe several things at once. For example, when assessing the airway and breathing you may note that the baby is very small or is restless. With practice, a complete triage (if no emergency treatment is needed) takes less than a minute.

SUMMARY

Assessment questions: Triage

Answer all the questions on this page, writing in the given spaces. If you have a problem, ask for help from one of the facilitators.

1. Define "triage".

2. When and where should triage take place?

3. Who should do triaging?

4. What do the letters A, B, C and D in "ABCD" stand for?

5. List the priority signs:

6. Put the actions in the right chronological order: what will you do first, what next, what after that, and so on, and what last?

- | | |
|----------------------|--|
| <input type="text"/> | Ask about head or neck trauma |
| <input type="text"/> | Call a senior health worker to see any emergency |
| <input type="text"/> | Have blood specimens taken for laboratory analysis |
| <input type="text"/> | Look for any priority signs |
| <input type="text"/> | Look for emergency signs |
| <input type="text"/> | Move on to the next patient |
| <input type="text"/> | Place priority patients at the front of the queue |
| <input type="text"/> | Start treatment of any emergency signs you find |

7. A three-year old girl is carried in her mother's arms wrapped in a blanket, in the queue. Her airway and breathing are OK. She has cold hands. Her capillary refill is 1.5 seconds. She is alert. Asked if the child has had diarrhoea, the mother answered "YES. Four loose stools per day". The skin pinch takes 3 seconds. How do you triage this child?

8. A four-year old male child was rushed in. He convulsed one hour ago. He is breathing fast but there is no cyanosis and no respiratory distress. He feels very hot, but responds quickly to questions. He has no diarrhoea or vomiting. How do you triage this child?

9. A one-year old had a seizure at home; then again outside the clinic. He became unconscious. His breathing sounds very wet and noisy and there is drooling coming from his mouth. He is looking blue. How do you triage this child?

10. A two-year old male is rushed to your clinic acutely convulsing. How do you triage this child?

11. What signs of malnutrition do you check during triage?

12. Where do you look for signs of severe wasting?

13. Below what age is a child always a priority?

14. What should you do if the child has a priority sign?